

GRID-CLOCK QUANTUM GRAVITY IV

A Finite Strict-Local Quantum Cellular Automaton for Emergent Spacetime, Einstein Gravity, Regular Black Holes, and Unitary Evaporation

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SCOPE NOTE. Version 3.2 preserves the one-law architecture and the E1-E9 verification baseline of v3.1, and adds the E10-E31 extension. The new work constructs a smooth-sector finite frame-holonomy dictionary for full directional curvature, identifies the finite transport resolution required by block refinement, introduces propagating two-polarization finite Q/P frame-wave registers, and connects the reconstructed curvature tensor to the spin-2 Hessian. In the Schwarzschild sector the resulting odd-parity curvature dynamics converges to the Regge-Wheeler potential; nondegenerate Darboux parity consistency then selects the Zerilli partner without fitting a separate polar coefficient. A factorized 18-phase all-register prototype now combines matter, load/capacity, frame/transport, reversible curvature caching, TT propagation, and curvature-controlled spin-2 phases with exact finite reversibility. Finite families of lapse- and shift-smear deformation circuits are also constructed, and their H-H, H-D, and D-D group commutators recover the intended physical-sector hypersurface-deformation relations under joint refinement. Version 3.2 still does not claim a completed quantum theory of gravity. The geometry-inclusive microscopic H-H circuit remains incomplete because the present integrated normal-deformation update does not yet contain the full geometry-momentum/curvature canonical action; the final frame/address representation, a full block-RG decoupling theorem, universal interacting matter, and uniqueness of the fundamental all-register update also remain open.

Abstract

Grid-Clock Quantum Gravity (GCC) is developed as a finite-information candidate architecture in which a fixed relational BCC graph and a universal microscopic update index are more fundamental than continuum spacetime. Version 3.2 extends the verification program through E31 while retaining a strict separation between exact finite statements, infrared derivations, constructive compatibility results, conditional completions, and open problems.

The flat doubled-Weyl BCC walk and the 32-dimensional curved-matter carrier are retained. The geometry sector is extended from scalar holonomy tests to a full directional-curvature reconstruction. General triads and Schwarzschild isotropic slices show approximately second-order recovery of the frame curvature tensor; a fixed binary-tetrahedral frame alphabet by itself is too coarse for weak links, while a finite gauge label tensored with a near-identity transport record admits the required refinement. Reversible block variables generate the transport resolution needed for the smooth limit and preserve gauge covariance.

A finite two-polarization Q/P frame-wave sector is then coupled to the same BCC geometry. Its microscopic matching scheduler has a massless isotropic Floquet cone with $O(a^2)$ error. Contracting the reconstructed three-dimensional curvature tensor with the physical spin-2 tensor representation produces the local transverse-traceless curvature Hessian directly. On a Schwarzschild slice its eigenvalues converge to $3 R_{\text{rhat_rhat}}$, yielding the Regge-Wheeler potential without fitting the curvature coefficient. Darboux parity consistency selects the corresponding Zerilli potential in the nondegenerate branch, and finite constraint projectors show how the reduced master variable can live in a finite Hilbert sector. These statements concern the verified Schwarzschild/linearized infrared branch; nonlinear Kerr, collapse, and strong-background stability remain separate completion problems.

The microscopic integration is also strengthened. An 18-phase factorized update combines geometry-controlled matter evolution and reciprocal backreaction, load/capacity refresh, finite frame/transport records, reversible loop-curvature caching, TT kinetic and BCC-gradient phases, and curvature-controlled spin-2 dynamics. Exact forward/inverse tests, coherent superpositions of discrete control sectors, and non-Abelian frame projectors show that the combined construction is a finite unitary existence prototype. Alternative symmetric orderings remain microscopically distinct while sharing the same low-energy limit, so existence is not promoted to uniqueness.

Keywords: quantum gravity; quantum cellular automaton; finite information; BCC lattice; finite Weyl algebra; frame holonomy; spin-2 curvature; Regge-Wheeler; Zerilli; hypersurface-deformation algebra; regular black holes; unitary evaporation.

Claim-status summary

Element	Status	Meaning
Fixed BCC graph and universal update index	Foundational postulates	Microscopic adjacency and update ordering are fundamental; continuum spacetime is emergent.
Finite local Hilbert spaces	Explicit quantum postulate	Geometry, matter-control, frame/load Link, scheduler, and work records are finite-dimensional.
Finite local capacity	Foundational completion postulate	Bounds localized energy and fixes the ultraviolet density scale.
U_GCC^full partitioned QCA	Constructive finite existence; uniqueness open	An 18-phase factorized all-register prototype combines matter/backreaction, load/capacity, frame/transport, curvature cache, and two TT polarizations with exact reversible composition. Different symmetric microscopic orderings remain possible.
Flat BCC/Weyl propagation	Exact reference sector	The four-component radius-one walk is exactly unitary and supplies the flat doubled-Weyl benchmark.
Universal curved-matter carrier	Constructive finite candidate	A 32-dimensional direction-register carrier remains unitary and radius-one under site-dependent anisotropic geometry; its flat branch is Weyl in the IR.
Matter backreaction	Exact finite identity + verified IR force	The same geometry-controlled unitary returns the conjugate geometry kick; anisotropic finite and low-quasienergy tests agree.
ADM/DeWitt gravity	Strong verified IR + circuit-level partial	$\lambda=1/2$ and the common structure function are selected in the tested sectors. Effective H-H/H-D/D-D and smeared-circuit tests close under joint refinement; the full geometry-inclusive microscopic H-H circuit is not yet complete.
Finite-register renormalization	Explicit collective and derivative-aware mechanism	Reversible block variables give $\eta_{\text{eff}}=\eta/B$ and $\zeta_{\text{eff}}=\zeta/B$; weak-link transport and derivative-entering registers require progressively finer IR resolution. Dynamic relative/frame decoupling has finite evidence but no full RG theorem.
Newton constant	Matched, not derived	Absolute normalization remains tied to the Planck calibration; stability and uniqueness under the full RG flow are open.
Frame-Link holonomy curvature	Constructive tensor reconstruction	BCC clovers reconstruct the full smooth frame-curvature tensor and directional Ricci data with approximately $O(a^2)$ error in tested general-triad and Schwarzschild backgrounds. The final microscopic frame/address representation and full nonlinear dictionary remain open.
Microscopic gauge / physical sector	Non-Abelian finite prototype + curvature-cache compatibility	Z3 and binary-tetrahedral 2T reductions preserve finite evolution and the collective mode; loop-curvature caches are exactly gauge invariant. The final GCC frame/address representation is not selected.
Planck-density core	Robust capacity result	Density bound and cube-root core scaling precede the detailed constitutive law.
Regular spherical profile	Conditional verified branch	The same bounded constitutive law connects the weak-load quadratic branch to a regular saturated core in the tested static sector.
Rank-two boundary algebra	Exact kinematic result	Orientation-independent outgoing support algebra is isomorphic to $M_2(\mathbb{C})$.
Binary horizon record	Conditional gauge reduction	Requires boundary-support minimality; no raw BCC edge is identified with a qubit.
Unitary evaporation and detailed balance	Explicit candidate sector	Finite-dimensional transfer and microcanonical normalization are explicit; greybody and exclusive

Element	Status	Meaning
		interacting amplitudes remain open.
Full interacting 3+1 theory	Open completion problem	All-register finite-unitary existence and substantial spin-2/HDA compatibility are now constructive, but unique microscopic ordering, full geometry-inclusive deformation algebra, final frame/address RG, universal interacting matter, collapse, nonlinear rotation, and all-sector RG remain unproved.
Propagating spin-2 sector	Constructive finite + verified IR	Two finite Q/P TT carriers, strict-local BCC matching gates, and curvature-controlled onsite phases produce a massless isotropic infrared wave sector; the finite low-energy quasienergy spectrum agrees with the split-step target.
Schwarzschild perturbation sector	Derived IR target + finite realization	Directional frame curvature gives the Regge-Wheeler odd potential; nondegenerate Darboux parity consistency selects the Zerilli partner. Finite constraint projectors and Q/P gates realize the reduced master sectors.
All-register integrated update	Exact finite existence prototype	Matter, backreaction, capacity/load, frame/transport, reversible curvature cache, TT propagation, and spin-2 curvature phases coexist in one autonomous finite composition with exact inverse tests.
Smeared deformation circuits	Strong / partial	Finite lapse/shift-dependent circuit families reproduce physical-sector H-H and geometry-aware H-D/D-D group-commutator targets under joint refinement. Full geometry-inclusive H-H awaits the complete geometry momentum/curvature action.

1. Introduction

The central problem addressed by GCC is not simply how to quantize a classical metric. It is whether quantum matter, relativistic causal propagation, Einstein gravity, finite strong-gravity behavior, and black-hole information transfer can be understood as different sectors of one finite local update law. This revision treats that question as the organizing principle of Paper IV.

The microscopic substrate is a fixed graph. The graph does not stretch, contract, or acquire continuum curvature as a fundamental object. What changes are finite records attached to nodes and Links: local frame/geometry information, matter amplitudes, capacity/load variables, and reversible scheduling data. Continuum distance, proper time, connection, metric, and curvature are relational observables reconstructed from these records. In this sense GCC differs in microscopic interpretation from a continuum theory in which the spacetime metric itself is fundamental, while still requiring general relativity as its smooth low-energy universality class.

The revision is built around a strict distinction between four logical levels. First, the microscopic QCA must be finite, unitary, autonomous, and causal. Second, a controlled smooth sector must recover relativistic matter and the ADM/DeWitt gravitational universality class. Third, spin-2 propagation and curvature must arise from the same finite frame/Link data rather than from an externally supplied wave equation. Fourth, strong-gravity and horizon statements must follow from the same finite records without a special horizon flag or a piecewise microscopic law. Version 3.2 also enforces a renormalization requirement: finite-register resolution must flow with lattice refinement whenever a continuum derivative would otherwise amplify cutoff errors.

$$U_{GCC}^{full} \rightarrow \left(\begin{array}{l} \text{Weyl/QFT and Lorentz symmetry} \\ \text{Einstein gravity in the smooth IR} \\ \text{regular strong gravity at saturation} \\ \text{unitary black-hole information transfer} \end{array} \right)$$

The rest of the paper constructs each arrow in this sector map and reports a verification stack through E31. The v3.2 extension makes the smooth frame-curvature, spin-2, all-register integration, and smeared-deformation interfaces constructive. It also isolates the remaining microscopic stopping point: the complete geometry-inclusive normal-deformation family, together with a final frame/address representation, full extra-mode decoupling, universal interacting matter, and a microscopic uniqueness principle.

PART I. ONE FINITE MICROSCOPIC LAW

2. Fixed graph, universal update, finite information, and capacity

GCC assumes a fixed relational BCC graph and a universal microscopic update label K . Coordinate labels and graph embeddings are auxiliary; only adjacency and relational records enter the microscopic law. The update label is an ordering variable, not directly an observable proper time. Local proper time and continuum lapse are emergent record-dependent quantities.

Finite information is implemented by finite-dimensional local Hilbert spaces. Finite energy capacity is a distinct postulate: a four-component propagation carrier does not imply four independent Planck-energy slots. This separation corrects the earlier Minimal Action Saturation interpretation and prevents Hilbert-space dimension from being misread as local energy capacity.

$$Z X = \omega X Z, \omega = e^{2\pi i/d}$$

Finite Weyl clock/shift pairs provide exact unitary record translations. Bounded Hermitian functions of these finite records are used for geometry, load, and local physical observables. The exact microscopic algebra is therefore finite; canonical commutators arise only in low-phase charts.

$$E_{cell} \leq E_P, \rho \leq \rho_P$$

When one BCC processing node is identified with one GCC capacity cell at Planck energy and Planck density, the relational processing scale satisfies the following calibration.

$$4 \ell_{proc}^3 = \ell_P^3, \ell_{proc} = 4^{-1/3} \ell_P, t_* = 4^{-1/3} t_P$$

This calibration fixes a branch of the microscopic scale dictionary. It does not turn the four propagation amplitudes into four independent energy capacities.

3. Full local Hilbert factorization

The minimal integrated architecture separates geometry, matter, capacity support, and reversible scheduling at nodes, and separates frame transport from scalar load on edges. This avoids the inconsistency that would result from using one Link phase simultaneously as a spin connection and as a strong-gravity load variable.

$$H = \otimes_x (H_{G_x} \otimes H_{M_x} \otimes H_{C_x} \otimes H_{W_x}) \otimes \otimes_e (H_{L_e^{frame}} \otimes H_{L_e^{load}})$$

The geometry register G contains a finite inverse-triad-like record and its finite Weyl conjugate refresh. The matter register M contains the strict-local propagation carrier and any additional internal species records. The Compute register C carries local support/capacity information. The work register W stores reversible scheduler, memory, and temporary holonomy accumulation data. Each edge has a frame Link, which compares local frames and transports matter, and a load Link, which carries the finite-Weyl scalar mismatch used by the Compute-Link saturation sector.

4. Autonomous strict-local QCA

The fundamental law is a discrete-time unitary, not the exponential of a local Hamiltonian. A finite-time exponential of a nearest-neighbor Hamiltonian generically develops longer-range tails, whereas the microscopic GCC postulate requires a hard one-neighborhood causal cone per update.

$$|\Psi_{K+1}\rangle = U_{GCC}^{full} |\Psi_K\rangle$$

To combine several local tasks without allowing information to propagate through multiple freshly updated neighborhoods inside one K -step, the QCA is partitioned and double-buffered. An internal scheduler register makes the update autonomous.

$$U_{GCC}^{full} = \sum_{s=0}^{R-1} |s+1\rangle \langle s|_W \otimes \left(\prod_{B \in P_s} u_B \right)$$

Within each scheduler phase the blocks in the partition are disjoint. Every local gate reads records from the current slice and writes the next slice; freshly written records are not reused as controls until the next update. Consequently a local Heisenberg observable can spread by at most one graph neighborhood per microscopic step.

$$\text{supp}(U^\dagger O_x U) \subseteq A_{N_1(x)}$$

The scheduler is not an external clock program. Its phase is a finite internal record shifted by the same autonomous law. Several microscopic ticks may form a convenient Floquet supercycle for infrared analysis, but the fundamental law remains one repeated QCA update.

5. Two levels of exactness and microscopic gauge

A finite commutative site algebra admits no nontrivial ordinary Leibniz derivation. GCC therefore does not demand the continuum infinitesimal hypersurface-deformation algebra as an exact operator identity at a fixed finite cutoff. Instead it assigns exactness to finite matrix identities, gauge-covariant update composition, and finite groupoid/path consistency. The ADM/DeWitt algebra is an emergent smooth-sector requirement.

Tangential deformation on the fixed graph is interpreted as relational address relabeling rather than literal graph motion. The finite triad record also has an internal frame redundancy because the metric is invariant under a local rotation of the triad. These redundancies define the microscopic gauge sector. A candidate physical subspace is obtained by projecting or averaging over the frame and address gauge actions.

$$P_{frame} = \int dg U_{frame}(g), P_{addr} = \frac{1}{|G_{addr}|} \sum_{\gamma \in G_{addr}} U_{addr}(\gamma)$$

$$P_{phys} = P_{addr} P_{frame}, H_{phys}^{micro} = P_{phys} H, [U_{GCC}^{full}, P_{phys}] = 0$$

If the full update is gauge covariant, the physical projector commutes with the QCA and the restricted evolution remains unitary. A gauge-singlet uniform vacuum shows that the projected space need not be empty. This is a construction route, not a classification theorem for the complete 3+1 physical Hilbert space.

PART II. MATTER AND EMERGENT RELATIVITY

6. Uniform vacuum: the strict-local BCC/Weyl sector

In a uniform, low-load geometry the full update must reduce to a homogeneous propagation QCA. The exact flat reference remains the four-component doubled-Clifford BCC walk. The v3.2 integrated curved completion, however, uses a 32-dimensional direction-register dilation $C^8_{\text{direction}}$ tensor C^4_{Clifford} as the universal local matter-carrier candidate. This distinction is forced by the verification audit: the 32-dimensional family supports arbitrary sitewise geometry with one fixed BCC shift, whereas a naive pointwise splice between the exact four-component stencil and the curved 32-dimensional scattering rule is not unitary.

$$\alpha_i = \sigma_x \otimes \sigma_i, P_i^\pm = \frac{I \pm \alpha_i}{2}, A_s = P_x^{s_x} P_y^{s_y} P_z^{s_z}$$

$$\psi_{K+1}(n) = \sum_{s \in \{\pm 1\}^3} A_s \psi_K(n-s)$$

The update is a single radius-one convolution. It is not interpreted as three sequential physical shifts. In Fourier space the same candidate can be written as a product of finite Clifford rotations.

$$U(q) = e^{-iq_x \alpha_x} e^{-iq_y \alpha_y} e^{-iq_z \alpha_z}$$

After a fixed carrier-basis rotation, the four-component reference walk decomposes into two Weyl sectors of opposite chirality. Published classifications of homogeneous, local, isotropic two-component walks in three dimensions select the BCC Cayley geometry and the two Weyl walks within the stated class. GCC therefore retains the four-component walk as an exact flat benchmark. The verification update does not yet prove that this walk is an

invariant four-dimensional subspace of the universal curved carrier; instead, the 32-dimensional family is required to reproduce the same Weyl principal cone in its low-wave-number flat sector.

7. Geometry-controlled propagation and the curved-space causal cone

The geometry record contains a finite triad-like variable. In a smooth semiclassical state it defines the spatial metric and inverse triad.

$$h_{ab} = E_a^i E_b^i, h^{ab} = e_i^a e_i^b$$

The local lapse and shift are read from the same geometry/scheduler record family. The matter QCA is not curved by moving the graph. Instead, the local scattering/coin and Link transport are conditioned on the geometry record. Its principal velocity matrix is chosen to approach the following smooth expression.

$$V^a[G] = N^a I + c N e_i^a \alpha^i$$

Using the Clifford anticommutation relations, the local dispersion satisfies the ADM null-cone relation.

$$(\omega - N^a k_a)^2 = c^2 N^2 h^{ab} k_a k_b$$

Thus a fixed microscopic graph can support an emergent curved-spacetime causal cone. The frame Links supply the covariant transport connection, so the low-energy one-particle generator has the Weyl/Dirac principal form.

$$H_M^{IR}[G, L] = \frac{1}{2} \left(N^a I + c N e_i^a \alpha^i, -i \hbar D_a \right) + H_{mass} + H_{int}$$

This establishes the kinematic curved-QFT branch conditionally: the exact finite QCA is fundamental, while the displayed differential operator is its smooth low-phase generator. In the E1 audit, a 3 x 3 x 3 inhomogeneous geometry pattern on the 32-dimensional carrier remained unitary with a relative residual 4.76×10^{-16} , propagated to exactly the eight BCC neighbors in one tick, and recovered the flat Weyl eigenphase with an $O(|k|^3)$ absolute error. Interacting many-body QFT still requires additional local species and interaction gates.

8. Unitary matter backreaction from the same coupling

A separate classical source rule is unnecessary at the architectural level. Let a finite geometry basis control a matter unitary.

$$C_{GM} = \sum_g |g\rangle \langle g| \otimes U_M(g)$$

Because the conjugate geometry update is a finite shift, conjugating that shift by the same controlled matter unitary gives an exact finite-dimensional identity.

$$C_{GM}^\dagger (X_G \otimes I) C_{GM} = X_G \Omega_G, \Omega_G = \sum_g |g\rangle \langle g| \otimes U_M(g+1)^\dagger U_M(g)$$

In a low-phase chart, where the finite shift is represented by the exponential of a conjugate momentum and the matter update by a short effective Hamiltonian step, the identity reduces to the ordinary Hamiltonian backreaction force.

$$\Delta p_A = -t \cdot \frac{\partial H_M}{\partial q^A} + O(t^2)$$

The key point is reciprocity: geometry controls matter propagation, and the same unitary necessarily updates the conjugate geometry record according to the geometry dependence of the matter dynamics. The E2 verification extends this statement to a 32-dimensional anisotropic carrier with shear, lapse-scale, and nonzero shift: the finite geometry-shift identity has residual 4.13×10^{-17} , a one-site geometry change produces a one-site relative kick, and the unitary Hellmann-Feynman relation between the conjugate connection and the derivative of the low quasienergy agrees to better than 2×10^{-8} relatively in the tested branches. A complete proof of universal coupling across all matter species is still required.

PART III. EMERGENT EINSTEIN GRAVITY

9. Finite geometry dynamics

The geometry update has a bounded kinetic part, a curvature part derived from frame Links, and an address/shift part. In the smooth chart the effective Hamiltonian is required to approach the most general local quadratic DeWitt kinetic form with a scalar-curvature potential.

$$H_G[N, N^a] = \int d^3x \left\{ N \left[\frac{A}{\sqrt{h}} (\pi^{ab} \pi_{ab} - \lambda \pi^2) - B \sqrt{h} R^{(3)} \right] + N^a D_a \right\}$$

At the microscopic level the momentum variables are bounded finite-Weyl functions rather than unbounded canonical operators. The displayed Hamiltonian is therefore an infrared generator of a finite unitary supercycle, not a fundamental continuous-time Hamiltonian. In the E4 reduced-sector construction, a bounded momentum function approaches the quadratic DeWitt kinetic term with an $\eta^{1.988}$ error law, reversible curvature compute-phase-uncompute returns the work register exactly to vacuum, and a three-phase autonomous Strang supercycle approaches its own bounded/cache effective generator with an $O(\tau^{2.020})$ error.

10. Closure selection of the DeWitt coefficient

The DeWitt trace coefficient is not left arbitrary. In the continuum bracket of two normal generators, the cross-term between the general quadratic kinetic term and the scalar-curvature potential produces the momentum constraint plus an additional trace-gradient term.

$$\{H_\lambda[N], H_\lambda[M]\} = D[\omega] + (1 - 2\lambda) \int d^3x (\nabla_a \pi) (N \nabla^a M - M \nabla^a N)$$

For generic states, if no additional trace constraint is introduced, the standard hypersurface-deformation algebra requires the coefficient of the extra term to vanish. Matching the normalization of the remaining momentum constraint also fixes the reciprocal kinetic/curvature product.

$$\lambda = \frac{1}{2}, AB = 1$$

The resulting target infrared Hamiltonian density is the ADM gravitational Hamiltonian with an effective Newton constant.

$$H_G = \frac{16\pi G_{eff}}{\sqrt{h}} \left(\pi^{ab} \pi_{ab} - \frac{1}{2} \pi^2 \right) - \frac{\sqrt{h}}{16\pi G_{eff}} R^{(3)}$$

The selection is conditional on the assumed local quadratic kinetic sector, the scalar-curvature potential, and the absence of an additional trace constraint. Within that sector it sharpens the earlier numerical observation that the closure root converges to the general-relativistic value.

11. Frame-Link holonomy and emergent spatial curvature

The fixed graph does not acquire microscopic curvature. Curvature is reconstructed from non-integrable frame transport. For each oriented BCC edge, the frame Link transforms covariantly under local frame rotations.

$$L_{x,s} \rightarrow G_x L_{x,s} G_{x+s}^\dagger, L_{x+s,-s} = L_{x,s}^\dagger$$

A minimal BCC four-edge loop provides a finite holonomy.

$$W_{x;s,t} = L_{x,s} L_{x+s,t} L_{x+t,s}^\dagger L_{x,t}^\dagger$$

In a smooth low-phase chart, the loop holonomy measures the curvature two-form through its area bivector.

$$W_{x;s,t} = \exp \left[-\frac{i}{4} \Sigma_{s,t}^{ab} F_{ab}^i \sigma_i + O(\ell_{proc}^3) \right]$$

Independent BCC loop bivectors span the three independent spatial two-form directions. Contracting the recovered frame curvature with the inverse triad gives the scalar curvature required by the ADM potential.

$$R^{[3]} = \epsilon_k^{ij} e_i^a e_j^b F_{ab}^k$$

A subtle locality point is essential. A node cannot instantaneously read all four Links of a loop whose opposite corner lies two graph steps away. GCC therefore uses a reversible work register to accumulate the loop product over several microscopic ticks. The E3-E4 verification makes this constructive: a controlled load-to-frame source gate is exactly unitary, a non-Abelian loop angle begins quadratically when the underlying Link angle is linear in load, the calibrated holonomy reproduces $f_g(\chi)$ to about 1.6×10^{-12} across a finite load register, and a four-tick loop product agrees with the direct holonomy to about 8×10^{-17} . The completed cache is then uncomputed at the supercycle boundary, avoiding a permanent work-mode degree of freedom.

Version 3.2 extends this construction from a scalar-curvature target to the full smooth frame-curvature tensor. General-triad BCC clovers recover the curvature two-form and directional Ricci data with approximately second-order error. On a Schwarzschild isotropic slice the scalar trace tends to zero while $R_{\text{rhat_rhat}}$ remains nonzero, and contracting the reconstructed Riemann/Ricci tensor with the physical sheet-TT representation yields the spin-2 curvature Hessian directly. A fixed 2T frame alphabet alone is too coarse for weak transport; the successful finite architecture separates the non-Abelian gauge label from a near-identity transport record whose effective resolution improves under block refinement.

12. Infrared hypersurface-deformation algebra

The geometric origin of the H-H structure function can be seen directly from the response of a unit normal to a normal deformation. Orthogonality to the tangents gives an inverse-metric-weighted lapse gradient.

$$\delta_N n = -h^{ab} (\partial_b N) e_a$$

The order difference of two normal deformations is therefore tangential and has the standard inverse-metric structure.

$$[\delta_N, \delta_M] X = e_a h^{ab} (N \partial_b M - M \partial_b N)$$

At the canonical level this is the target normal-normal bracket.

$$\{H[N], H[M]\} = D[h^{ab} (N \partial_b M - M \partial_b N)]$$

The existing numerical audits support the distinction between full-cutoff and smooth-sector closure. The earlier low-mode results below remain useful as a regulator audit. Version 3.1 established a full three-dimensional canonical test with six h_{ab} and six π^{ab} components in which bounded finite-phase momentum and curvature maps are inserted into the V4 Hamiltonian. With no deformation, H-H, H-D, and D-D residuals refine approximately as $a^{1.946}$, $a^{1.927}$, and $a^{1.983}$. If the finite-register deformation parameters are refined jointly as $\eta_p = \zeta_R = 0.30 a$, the fitted powers are $a^{1.969}$, $a^{2.680}$, and $a^{1.983}$, respectively. The DeWitt scan continues to select $\lambda = 1/2$.

epsilon	Raw H-H diamond	Residual after fitted shift	Fitted coefficient
0.040	3.17×10^{-4}	2.52×10^{-6}	0.99552
0.020	7.93×10^{-5}	3.01×10^{-7}	0.99723
0.010	1.98×10^{-5}	3.49×10^{-8}	0.99810
0.005	4.96×10^{-6}	4.16×10^{-9}	0.99853

Sites	a	Low-mode H-D residual	Low-mode D-D residual
32	0.19635	0.11249	0.17904
64	0.09817	0.02875	0.04623
128	0.04909	0.007227	0.01165
256	0.02454	0.001809	0.002918

A crucial negative control accompanies this result. If the bounded deformation is held fixed while only the spatial lattice is refined, the H-H residual does not converge to the ADM target; at $N=40$ the tested fixed-deformation

branch remains near 2.88×10^{-2} . Thus finite-register corrections must themselves flow under refinement. The E6 block construction below supplies a concrete collective-mode mechanism for that flow, but these results are still evidence for the required infrared universality class rather than a proof that every nonlinear many-body sector of the full QCA is anomaly-free.

Version 3.2 repeats the closure test after matter and two physical TT sectors are integrated. With a common inverse-metric structure function, the H-H error again converges approximately as dx^2 , while a 10 percent TT gradient-normalization mismatch leaves a finite closure floor. Finite $H[N]$ and $D[xi]$ circuit families then reproduce physical-sector H-H group commutators and geometry-aware H-D/D-D commutators under joint spatial, geometry-register, and smearing-register refinement. The full-state H-H circuit remains open because the current normal update does not yet evolve the complete geometry canonical pair.

13. Einstein-Hilbert reconstruction and normalization status

If the geometry, matter, gauge, and physical spin-2 sectors described above share the same smooth lapse, shift, and inverse-metric structure function, if the complete geometry-inclusive H-H/H-D/D-D algebra closes in the refined physical sector, and if no additional low-energy gravitational degree of freedom survives, the effective action takes the Einstein-Hilbert plus matter form with higher-derivative ultraviolet corrections. Version 3.2 strengthens this route by constructing the smooth directional-curvature and spin-2 sectors, an all-register finite-unitary existence prototype, and explicit smeared deformation circuits. The remaining covariance gap is no longer hidden in the matter or TT sector: it is localized to the full geometry canonical action in the normal-deformation family, while extra frame-mode decoupling remains a separate dynamical requirement.

$$S_{eff} = \frac{c^3}{16\pi G_{eff}} \int d^4x \sqrt{-g} R + S_{matter} + \Delta S_{UV}$$

The relative normalization between the DeWitt kinetic term and curvature term is constrained by closure. The absolute value of the effective Newton constant is not independently derived in the present construction and remains matched to the GCC Planck calibration. The E6 block audit gives an explicit finite-Weyl collective flow $\eta_{eff} = \eta/B$ and $\zeta_{eff} = \zeta/B$, showing how bounded microscopic corrections can weaken under coarse graining, but it does not yet establish the stability or uniqueness of the absolute Newton normalization in the full interacting theory.

PART IV. FINITE CAPACITY AND REGULAR STRONG GRAVITY

14. Capacity saturation and robust core scaling

No elementary capacity cell is required to represent arbitrarily large localized energy. The density bound alone implies a minimum volume for a fixed enclosed mass. In spherical symmetry this gives the robust cube-root core scale.

$$r_c = \left(\frac{3M}{4\pi\rho_p} \right)^{1/3} \propto M^{1/3}$$

This scaling does not require a Hayward metric, a particular Link harmonic, or an equal Compute-Link load assumption. It follows from finite density, three-dimensional volume counting, and finite total mass. The detailed radial profile requires an additional constitutive closure.

15. Explicit Compute-Link circuit and the correlated radial equation

The earlier quadratic strong-gravity branch can be realized by two required local support roles: Compute and Link. The revised architecture retains the explicit reversible correlation circuit rather than replacing its quantum correlations by an arbitrary radial interpolation. Using the mean enclosed density as the coarse load gives the dimensionless variable below.

$$\chi(r) = \frac{3m(r)}{4\pi\rho_p r^3}$$

After tracing only reversible work registers, the explicit Compute-Link circuit gives the normalized density response

$$f_g(\chi) = \frac{\rho}{\rho_p} = \chi^2 + \chi(1 - \chi) \sin^2(g \chi^2)$$

The response is bounded and has exact vacuum and saturation endpoints.

$$0 \leq f_g(\chi) \leq 1, f_g(0) = 0, f_g(1) = 1$$

Combining mass conservation with the definition of the mean-density load yields a closed radial equation for the full correlated circuit.

$$r \frac{d\chi}{dr} = -3 \chi(1 - \chi) \cos^2(g \chi^2)$$

This equation improves the logical status of the strong-gravity branch. The radial dependence is no longer freely chosen once the circuit response is specified. The remaining completion choice is the microscopic identification between the scalar Link load and the coarse spherical load.

16. Hayward limit, correlated deformations, and exterior behavior

In the uncorrelated limit the radial equation integrates exactly.

$$g=0: \chi(r) = \frac{a^3}{r^3 + a^3}, m(r) = \frac{M r^3}{r^3 + a^3}, a^3 = \frac{3M}{4\pi\rho_p}$$

This is the Hayward-type mass profile used in the earlier revision. For nonzero circuit correlation the exact Hayward form is deformed, but the density cap and central saturation remain intact. Near the vacuum the correlation correction enters only at fifth order in the load; therefore the leading $p=2$ exterior correction remains quartic in inverse radius, while the new correlation-dependent correction appears at much higher inverse power. The detailed coefficient is microscopic-model dependent.

At the center the load approaches unity and the density approaches the Planck value. The corresponding static spherical geometry is de Sitter-like rather than singular. This establishes a regular strong-gravity branch conditionally on the chosen local constitutive dictionary; it does not prove dynamical collapse into that branch or stability of an inner horizon.

17. Frame curvature and scalar load must be distinct Link factors

A key consistency correction is that the frame-holonomy phase of Section 11 cannot be identified directly with the scalar finite-Weyl load phase. In the weak-field Einstein sector, curvature is proportional to energy density; in the minimal two-channel branch the density is quadratic in the load. Therefore the curvature holonomy begins quadratically in the scalar load, not linearly.

$$H_L = H_{L^{frame}} \otimes H_{L^{load}}$$

$$\theta_{curvature} \propto f_g(\chi) = \chi^2 + O(\chi^5)$$

The two Link factors remain parts of one neighboring-record architecture. The frame factor transports local orientation and reconstructs curvature; the scalar factor carries the bounded capacity mismatch. In the E3 source-gate prototype the load register controls two noncommuting $SU(2)$ frame rotations; their loop holonomy is quadratic at weak load and is calibrated once so that the normalized curvature response equals $f_g(\chi)$ from vacuum to saturation. The controlled source gate is unitary to machine precision and its reciprocal load-shift identity is exact. This makes the separation constructive while leaving the unique microscopic source gate and the full three-dimensional frame-to-curvature dictionary open.

PART V. HORIZONS AND UNITARY INFORMATION TRANSFER

18. The horizon is an emergent cut, not a microscopic defect

The full QCA contains no horizon flag. Event-horizon membership is a global causal property, while the local curvature at the horizon of a very massive black hole can be small. A special surface coupling would therefore introduce a nonlocal retuning signal. In the minimal GCC branch, the same local update acts everywhere; a horizon is identified only after a semiclassical state is partitioned into inside and outside causal sectors.

19. Orientation-independent rank-two boundary algebra

Raw BCC edge counting is regulator dependent and fails as a universal horizon count. The correct kinematic object is the normal-current split of the same four-component bulk carrier. For any physical unit normal, the Clifford algebra gives two outgoing and two ingoing eigenchannels.

$$P_{out} = \frac{I + \alpha_{\hat{n}}}{2}, P_{in} = \frac{I - \alpha_{\hat{n}}}{2}, \text{rank } P_{out} = \text{rank } P_{in} = 2$$

The outgoing support algebra is therefore

$$A_H = P_{out} M_4(C) P_{out} \simeq M_2(C)$$

This result is orientation and frame independent because local frame rotations act by unitary conjugation. It is an exact kinematic statement about the bulk carrier. It is not, by itself, a theorem that the full many-body horizon Hilbert space contains no additional independent memory.

20. Boundary support minimality, counting, and area matching

The remaining counting assumption can be sharpened as boundary-support minimality: after the inside/outside split and gauge reduction, independent horizon memory is exhausted by the outgoing support algebra of the universal bulk carrier; species and polarization labels are scattering channels rather than independent surface-memory factors. Under this condition the minimal irreducible boundary record is binary.

For a closed connected horizon, one global binary consistency condition reduces the raw count by one factor of two.

$$\Omega_F = 2^{F-1}, S_H = k_B (F - 1) \ln 2$$

Finite-Weyl kinematics fixes the entropy increment of one independent record but not its physical area. The conversion is fixed by matching to the required semiclassical stationary-horizon first law and Hawking temperature. One binary record then corresponds to

$$\Delta A = 4 \ln 2 \ell_p^2$$

The area coefficient is therefore a micro-IR matching result, not a pure finite-state counting theorem.

21. Discrete horizon tower and exact finite information transfer

Using the regular extremal geometry as an information-free offset, the horizon information sectors may be organized by the number of independent binary records.

$$A_n = A_{ext} + 4 n \ln 2 \ell_p^2, S_n = k_B n \ln 2$$

The crucial dimension identity for a transition that transfers q records to radiation is exact.

$$\dim H_H(n) = 2^n = 2^{n-q} 2^q = \dim [H_H(n-q) \otimes H_R(q)]$$

Therefore each energy-compatible sector admits a unitary isomorphism between the initial horizon space and the smaller horizon space tensored with the emitted record space. For one record, the explicit map used in Paper IV is

$$\begin{aligned} |g_n; b_1, \dots, b_n\rangle |0\rangle_R &\mapsto |g_{n-1}; b_1 \oplus b_n, b_2, \dots, b_{n-1}\rangle |b_n, \epsilon_n\rangle_R \\ M_n c^2 &= M_{n-1} c^2 + \epsilon_n \end{aligned}$$

The map is bijective and extends to an exact finite-dimensional unitary. Repeated application transfers progenitor boundary information to radiation/history correlations without requiring the final geometric endpoint to carry a large information remnant. In the full theory this map is not a special horizon gate; it is the effective block of the same universal QCA restricted to the corresponding black-hole/radiation sector.

22. Microcanonical reciprocity and inclusive Hawking normalization

Because the microscopic law is reversible, the inclusive emission and absorption matrix-element sums differ only by the number of initial black-hole microstates. For a transition removing q independent records,

$$\frac{P_{emit}}{P_{abs}} = \frac{\Omega_{n-q}}{\Omega_n} = 2^{-q}$$

If the no-boundary-defect branch is used, the absorption probability is the greybody probability of the same exterior QCA. The inclusive emission probability is therefore

$$P_{emit}(n \rightarrow n-q; \alpha) = 2^{-q} T_\alpha(\omega_{nq})$$

No independent phenomenological horizon-source constant is required. The remaining unknowns are the exterior QCA scattering matrix, mode density and group velocity, and exclusive many-body amplitudes. In a smooth normalization the inclusive occupation takes the usual Bose/Fermi entropy form, with the QCA flux measure replacing the continuum one when necessary.

$$\frac{d^2 N}{dt d\omega} = \sum_\alpha \Phi_{QCA, \alpha}(\omega) \frac{T_\alpha(\omega)}{\exp[\Delta S_{BH}(E, \omega)/k_B] \mp 1}$$

23. Sector-centroid gaps, Page turnover, and quasienergy

The area tower gives a sector-centroid energy gap that approaches the Hawking temperature times one binary entropy unit for a large Schwarzschild black hole.

$$\epsilon_n \simeq k_B T_H \ln 2, \frac{\epsilon_n}{k_B T_H \ln 2} = 2n - 2\sqrt{n(n-1)}$$

The tower does not require exact degeneracy of all microstates at fixed n . Interactions in the full QCA may split the exponentially many microstates into a dense band, so a resolvable line spectrum is conditional rather than robust. The more robust prediction is the finite-sector entropy correction to detailed balance.

For a typical initially scrambled state, unitary record transfer gives the standard Page turnover.

$$S_R(n) \simeq k_B \ln 2 \min(n_0 - n, n), M_{Page} = \frac{M_0}{\sqrt{2}}, \frac{t_{Page}}{\tau} = 1 - \frac{1}{2\sqrt{2}}$$

Because the microscopic dynamics is discrete-time, exact conservation is most naturally formulated in terms of QCA quasienergy. For an R-tick autonomous supercycle,

$$U = (U_{GCC}^{full})^R, H_{eff} = \frac{i\hbar}{Rt_*} \log U$$

$$\varepsilon \sim \varepsilon + \frac{2\pi\hbar}{Rt_*}$$

In the low-phase branch this quasienergy is identified with ordinary physical energy. The blockwise mass-gap relation should therefore be read as an exact sector matching within the chosen low-phase energy branch, not as a claim that an unbounded Hamiltonian is fundamental.

PART VI. INTEGRATED AUDIT AND VERIFICATION PROGRAM

24. One-law sector map

The construction is now organized around one microscopic architecture rather than a collection of independent models. The following table summarizes the sector conditions and the corresponding effective theory.

Sector / limit	Dominant records and update	Effective result	Current status
Uniform vacuum; low wave number	Exact 4D BCC reference; 32D universal candidate in flat geometry	Doubled Weyl cone	Exact reference + verified IR-equivalent universal branch
Smooth geometry; low load	32D direction coin + fixed BCC shift + frame transport	Curved Weyl/Dirac principal dynamics	Constructive finite carrier; IR verified
Dynamical smooth geometry	Same geometry-controlled matter unitary	Reciprocal matter backreaction	Exact finite identity; anisotropic force verified
Finite geometry supercycle	Bounded finite-Weyl kinetic + reversible loop/cache + scheduler	ADM/DeWitt effective generator	Constructive reduced sector; $O(\tau^{\wedge 2})$

Sector / limit	Dominant records and update	Effective result	Current status
Refined 3D sector	Joint lattice/register refinement	H-H/H-D/D-D $\rightarrow$ ADM	Verified IR; fixed-deformation negative control
Coarse-grained physical sector	Reversible block variables + relative-mode dynamics + gauge projection	$\eta, \zeta \rightarrow 1/B$; collective low mode	Exact collective flow + finite interacting/gauge prototypes
High capacity load	Compute + load-Link + frame source circuit	Planck density bound; regular core	Robust bound + verified static constitutive branch
Semiclassical inside/outside cut	Normal-current projection of bulk carrier	Rank-two boundary algebra	Exact kinematic result
Evaporating BH sector	Same QCA restricted to horizon/radiation blocks	Unitary information transfer + microcanonical emission	Explicit finite sector; scattering incomplete
Smooth spin-2 sector	Finite TT Q/P carriers + BCC matching propagation + frame-curvature Hessian	Massless TT cone; Regge-Wheeler/Zerilli Schwarzschild branch	Constructive finite carrier + verified linear IR
Integrated microscopic prototype	Matter/backreaction + capacity/load + frame/transport + curvature cache + TT sectors	One factorized all-register reversible supercycle	Exact finite existence prototype; ordering non-unique
Smeared deformation sector	Finite lapse/shift controls + $H[N]$, $D[xi]$ circuit families	Physical H-H and geometry-aware H-D/D-D HDA relations	Strong / partial; full geometry H-H open

25. What is exact, derived, conditional, and open

Exact finite statements now include the finite Weyl algebra, unitarity of the chosen local gates, strict partitioned causality, geometry-controlled backreaction identity, reversible loop/cache compute-uncompute, collective block changes of variables, non-Abelian frame projectors, finite TT Q/P gates, the all-register 18-phase forward/inverse composition, gauge-invariant finite loop-curvature caching, finite constraint projectors, and exact finite symplectic geometry-momentum shears. These are matrix or permutation statements at finite register dimension and should not be confused with continuum uniqueness.

Derived and numerically verified infrared statements now include the ADM causal cone, DeWitt one-half selection, full canonical H-H/H-D/D-D convergence under joint lattice/register refinement, approximately second-order frame-curvature reconstruction, a massless two-polarization BCC TT cone, the Schwarzschild spin-2 curvature coefficient, Regge-Wheeler/Zerilli master-sector recovery, and circuit-level smeared deformation relations on the verified physical sectors. The E29/E31 results strengthen compatibility but do not turn the present update into a theorem that every nonlinear many-body sector is anomaly-free.

Conditional completions include promotion of the 32-dimensional matter carrier to the universal interacting carrier, completion of the microscopic geometry canonical pair and curvature momentum update in the smeared normal-deformation family, dynamic decoupling of every relative and physical frame excitation, controlled coarse graining to the full Einstein-Hilbert action, the absolute Newton normalization, the scalar Link-load dictionary beyond the tested sources, boundary-support minimality, and the embedding of exclusive evaporation amplitudes in the same universal QCA. The Schwarzschild linearized spin-2 chain does not by itself establish nonlinear Kerr or collapse dynamics.

Open problems have narrowed but remain decisive. The v3.2 construction demonstrates an all-register finite-unitary prototype but not a unique fundamental U_{GCC}^{full} . The final finite frame/address representation is not uniquely selected, and the full interacting block RG has not proved that all extra physical frame/relative modes decouple. Most importantly for covariance, finite smeared $H[N]$ and $D[xi]$ families now exist, yet the current full-state H-H circuit fails for a specific architectural reason: the normal-deformation update does not yet evolve the spatial geometry through its complete conjugate momentum and curvature sector. Universal interacting matter, absolute Newton normalization, nonlinear collapse/rotation, inner-horizon stability, greybody dynamics, and exclusive Page evolution remain open.

26. Falsifiability and consistency conditions

- No candidate microscopic update is acceptable if one K-step propagates information beyond the declared graph neighborhood.
- The refined low-energy sector must recover the complete H-H, H-D, and D-D algebra; success in only one bracket is insufficient.

- The same geometry record that controls matter propagation must generate compatible backreaction and must not require species-dependent retuning that violates universal coupling.
- The physical gauge-reduced Hilbert space must contain nontrivial excitations and be invariant under the full update.
- The refinement/renormalization map must suppress lattice Lorentz and diffeomorphism violations at observable scales.
- The finite-capacity branch must preserve the density bound and remain dynamically stable under collapse and rotation.
- The scalar Link load dictionary and harmonic assumptions must be derived, independently motivated, or empirically constrained.
- The boundary rank-two sector and information-transfer map must survive the full many-body construction without an additional nonlocal horizon flag.
- Once the exterior absorption problem is calculated, emission must obey the microcanonical reciprocity normalization rather than require a new fitted horizon coupling.

27. Numerical verification status and extension stack

The computational program is now extended from the V1-V6 and E1-E9 baseline through E31. The extension is organized as a sequence of constructive attempts and negative controls. PASS denotes only the stated finite or infrared test, not completion of quantum gravity; STRONG / PARTIAL denotes a verified mechanism whose microscopic uniqueness or all-sector closure remains open.

$$\varepsilon_U = \| U^\dagger U - I \|, \varepsilon_{caus}(r>1) = \| \Pi_{r>1}(U^\dagger O_x U) \|$$

The V1-V6 and E1-E9 stack established the exact flat BCC reference, curved 32-dimensional matter carrier, reciprocal backreaction, finite load-to-frame source, bounded geometry supercycle, joint-refinement HDA target, block coarse graining, and non-Abelian gauge prototype. Version 3.2 then targets the specific gaps identified by that stack: full directional curvature, propagating finite spin-2 carriers, black-hole perturbation sectors, all-register integration, and finite smeared deformation circuits.

$$\varepsilon_{HDA}(a, k_{max}) \rightarrow 0, \varepsilon_{Lorentz}(k \ell_{proc}) \rightarrow 0$$

E10-E13 construct the smooth finite frame-curvature and universality bridge; E14-E18 audit strong-gravity observability, angular momentum, frame dragging, and spin-squared response; E19-E22 build the unitary no-hair and finite TT radiation mechanisms; E23-E27 connect curved finite wave gates, finite constraint reduction, and directional frame curvature to the Regge-Wheeler/Zerilli sectors; E28 derives the spin-2 curvature Hessian from the reconstructed tensor; E29 assembles an all-register finite-unitary existence prototype; E30-E31 test integrated and circuit-level HDA structure and isolate the missing full geometry normal-deformation action.

Stage	Primary result	Status
V1. Flat matrix law	Exact 4D BCC radius-one unitary and stencil identity	PASS
V2. Isotropic geometry control	Finite controlled unitary, locality, reciprocal backreaction	PASS
V3. General ADM principal symbol	32D direction dilation; anisotropic triad + shift; $O(k^3)$ phase error	PASS
V4. 3D ADM/HDA target	$\lambda=1/2$, $AB=1$; H-H/H-D/D-D refine approximately as a^2	PASS (IR)
V5. Static strong gravity	One bounded $f_g(\chi)$ from vacuum to saturation; regular center	PASS (static)
V6. Integrated common-law audit	No sector contradiction; one explicit all-register U_{GCC}^{full} still absent	CONDITIONAL
E1. Common matter carrier	Universal 32D sitewise-geometry family works; naive 4D/32D splice fails	PARTIAL
E2. Anisotropic backreaction	Exact finite identity + local kick + low-quasienergy force agreement	PASS
E3. Load-to-frame source	Non-Abelian holonomy, quadratic weak load, fifth-order correlation	PASS
E4. Autonomous geometry cycle	Reversible loop/cache, bounded kinetic, $O(\tau^2)$ Floquet generator	PARTIAL
E5. Bounded full 3D HDA	Joint lattice/register refinement closes; fixed deformation leaves H-H floor	STRONG / PARTIAL

Stage	Primary result	Status
E6. Block coarse graining	Exact collective $\eta, \zeta \rightarrow 1/B$; cache resolution $\rightarrow 1/B$	PARTIAL
E7. Dynamic collective sector	Local stiffness pushes relative mode above collective low sector	PARTIAL
E8. Finite gauge prototype	Gauge projection preserves collective sector and finite unitary	PARTIAL
E9. Non-Abelian frame prototype	2T physical projector and gauge compatibility; extra frame modes/dictionary open	PARTIAL
E10. Finite frame curvature	General-triad BCC clovers; hybrid 2T gauge x near-identity transport; RG resolution; $O(a^2)$ scalar/tensor recovery	PASS / constructive IR
E11. Extra-mode decoupling	Global-shift protection and BCC/frame spectral separation; collective sector becomes parametrically low	STRONG / PARTIAL
E12. Autonomous integrated QCA	Finite matter/load/frame prototype exists; corrected frame-Link representation; microscopic choices are non-unique	STRONG / PARTIAL
E13. Universality audit	Relevant IR data separated from $O(a^2)$ lattice corrections; strong-gravity constitutive freedom remains	STRONG / PARTIAL
E14. Static BH observability	Planck-core exterior shifts scale beyond practical shadow/ringdown sensitivity; echoes require an added reflector	PASS (audit)
E15. Collapse and rotation localization	Density cap does not select bounce; core is causally hidden after horizon formation; astrophysical spin must be extended	PASS (audit) / OPEN dynamics
E16. Finite angular momentum	Exact finite angular-momentum exchange and global BCC conservation; weak frame dragging converges as $O(a^2)$	PASS (constructive weak rotation)
E17. Kerr audit	Weak frame-drag normalization matches $\kappa=2$ in the IR; spin-squared quadrupole amplitude and nonlinear extremality are not fixed	STRONG / PARTIAL
E18. Spin-squared source	Exact finite STF source/backreaction shapes; horizon regularity supplies an IR Kerr selector, not a microscopic proof	STRONG / PARTIAL
E19. Unitary no-hair	Local anomalous quadrupole can radiate into finite wave channels; no-hair requires absence of stable bound modes	STRONG / PARTIAL
E20. Radiation-spectrum audit	Existing E12 lacks a propagating frame-wave gate; BCC TT target and bound-mode criterion identified explicitly	STRONG / PARTIAL
E21. Finite frame-wave gate	Exact finite Q/P TT gates and BCC scheduler; actual Floquet cone $\omega \rightarrow c k $ with $O(a^2)$ error	STRONG / PARTIAL
E22. Quadrupole-source reduction	No independent defect oscillator is required in the minimal architecture; source shifts equilibrium while TT perturbations propagate	STRONG / PARTIAL
E23. Curved spin-2 target	Naive curved scalar wave fails Zerilli; finite reduced Zerilli gate and local load representation succeed	STRONG / PARTIAL
E24. Finite constraint reduction	Four finite even-sector pairs reduce to one physical pair; finite projector and Schur-complement mechanism are explicit	STRONG / PARTIAL
E25. Directional-curvature coefficient	$R^{(3)}=0$ Schwarzschild slice still carries traceless Ricci; exact local target is lapse + directional curvature; coefficient family not yet unique	STRONG / PARTIAL
E26. Parity-Darboux selector	Nondegenerate Darboux consistency selects Regge-Wheeler/Zerilli coefficient family; finite intertwiner converges	STRONG / PARTIAL
E27. Odd curvature derivation	Spin-2 curvature algebra fixes 3 $R_{\text{rhat}}_{\text{rhat}}$; BCC holonomy gives $k=4$ and Regge-Wheeler, then Darboux fixes Zerilli	STRONG / PARTIAL

Stage	Primary result	Status
E28. Spin-2 Hessian	Full reconstructed curvature tensor directly yields TT Hessian eigenvalues $\rightarrow 3$ $R_{\text{rhat_rhat}}$; exact finite two-polarization gate	STRONG / PARTIAL
E29. All-register integration	18-phase matter+geometry+frame+cache+TT composition is exactly reversible; coherent controls and projectors remain compatible	PASS existence / uniqueness OPEN
E30. Integrated HDA	Matter+TT H-H and geometry-aware H-D/D-D converge under joint refinement; frozen geometry leaves an H-D floor	STRONG / PARTIAL
E31. Smeared deformation circuits	Finite $H[N], D[xi]$ circuit families reproduce physical H-H and H-D/D-D group commutators; full H-H exposes missing geometry momentum action	STRONG / PARTIAL

27.1 Finite-register renormalization and the E5-E7 bridge

The E5 negative control fixes an important consistency condition. Holding the finite deformation scales fixed while $a \rightarrow 0$ leaves a nonzero H-H floor; at $N=40$ the tested $\eta_p = \zeta_R = 0.25$ branch remains near 2.884×10^{-2} . By contrast, the joint choice $\eta_p = \zeta_R = 0.30$ gives H-H, H-D, and D-D refinement powers 1.969, 2.680, and 1.983. At fixed $N=24$, the excess bracket errors from the kinetic and curvature finite-phase deformations are each approximately quadratic in their respective deformation parameters.

$$\eta_{\text{eff}} = \eta_{\text{micro}} / B, \quad \zeta_{\text{eff}} = \zeta_{\text{micro}} / B$$

E6 supplies the corresponding microscopic collective-mode mechanism. A reversible finite-Weyl block transform preserves the discrete canonical pairing exactly, and in the retained collective mode both bounded deformation scales flow as $1/B$, while the normalized finite cache resolution improves as B^{-1} . E7 then relaxes the hand-imposed relative-mode condition: one fixed nearest-neighbor stiffness separates a low collective excitation from relative excitations, and the collective gap approaches the E6 target as the finite-register dimension and block size are increased. This is evidence for an RG mechanism, not yet a proof that the full interacting QCA dynamically closes on the same local architecture.

27.2 Gauge-reduced finite frame verification

E8 first demonstrated the compatibility of the collective mechanism with an explicit finite gauge reduction in a Z3 plaquette prototype. E9 replaces that abelian example with the binary-tetrahedral group $2T$, a non-Abelian finite subgroup of $SU(2)$ with order 24. Group averaging gives a seven-dimensional conjugation-invariant physical frame sector, with projector residual 6.50×10^{-16} ; the tested frame Hamiltonian commutes with the gauge action exactly, and the coupled scalar/frame projector commutator is 4.33×10^{-16} .

$$[P_{\text{phys}}, U] = 0$$

The collective scalar geometry mode survives this non-Abelian reduction. At local scalar dimension $d=11$ the collective gap is 0.204356 against the E6 target 0.199933, while the first relative gap is 0.768415. A finite partitioned evolution remains unitary at roundoff and its physical quasienergy error scales as $\tau^2.00042$. The same audit exposes a remaining no-extra-mode problem: the first pure physical frame gap in the main $2T$ coefficients is only 1.687 times the collective target. Increasing the frame magnetic coefficient can enlarge the gap, but that coefficient is not uniquely derived, so parameter tuning is not counted as a completion.

27.3 Current microscopic stopping point

The verification update therefore changes the location of the main open problem. Spatially varying BCC frame holonomies now reconstruct the smooth full curvature tensor in explicit general-triad and Schwarzschild tests, and that tensor drives a finite physical spin-2 sector. An all-register finite-unitary supercycle and finite smeared deformation-circuit families also exist. What is not yet derived is the complete geometry-inclusive normal-deformation action whose group commutator closes the full H-H relation on the spatial geometry itself. The final completion must additionally select the microscopic frame/address representation, prove dynamic decoupling of extra frame/relative modes under the real block RG, establish universal interacting matter and Newton normalization, and extend the verified linear Schwarzschild sector to nonlinear collapse and rotation.

28. Version 3.2 extended-verification overview

Version 3.2 does not alter the finite-information postulates or the strong-gravity/horizon architecture of the earlier revision. Its main purpose is to attack the previously explicit geometric seams. The extension starts from the finite frame Links themselves, asks which finite representations can encode weak smooth transport, reconstructs the full frame curvature tensor rather than only a scalar load, introduces finite physical spin-2 carriers, and then follows those carriers through Schwarzschild perturbation theory, an all-register finite update, and smeared deformation circuits.

The resulting evidence is substantially stronger than a collection of continuum fits. Several new layers are exact finite statements: reversible loop-curvature caches, TT Q/P gates, finite constraint projectors, all-register forward/inverse compositions, coherent-control compatibility, and finite lapse/shift-dependent deformation circuits. Other layers are infrared derivations with explicit refinement tests: directional curvature, Regge-Wheeler/Zerilli recovery, spin-2 Floquet dispersion, and integrated HDA closure. Negative controls are retained whenever they identify a necessary condition, including failure of a fixed 2T weak-link alphabet, fixed register resolutions, propagation-speed mismatch, naive curved-scalar substitution for Zerilli, and frozen-geometry H-D closure.

The sections below summarize E10-E31 as one verification chain. The central stopping point is deliberately preserved: all-register finite-unitary existence is now constructive, but a unique microscopic law and the complete geometry-inclusive smeared deformation algebra are not yet derived. This distinction is the main claim-status boundary of v3.2.

PART VII. EXTENDED VERIFICATION: CURVATURE, SPIN-2 DYNAMICS, AND SMEARED DEFORMATIONS

29. Finite frame curvature, renormalization, and universality (E10-E13)

E10 replaces the earlier scalar-holonomy sketch by a constructive smooth-sector curvature reconstruction. Four BCC directions generate six independent direction pairs whose area bivectors span the spatial two-form space. Path-ordered frame Links and four-edge clovers are reconstructed at a common basepoint, and least-squares inversion recovers the frame curvature two-form. In conformally flat tests the scalar-curvature relative error falls from 6.82×10^{-2} at $a=0.08$ to 1.10×10^{-3} at $a=0.01$ with fitted power 1.988. A general non-diagonal triad gives the same approximately second-order behavior. Local $SO(3)$ frame rotations preserve the scalar to roundoff.

A finite-representation negative control is essential. A fixed binary-tetrahedral 2T/T frame alphabet has a minimum nontrivial rotation of 120 degrees and therefore cannot by itself represent arbitrarily weak smooth Links. The successful finite representation is a product architecture: a finite non-Abelian endpoint gauge label together with a finite near-identity transport record. The transport resolution must refine faster than the area scale. In the constructive block map, a b^3 block averages the collective transport increment as $1/B$, giving an effective link-angle resolution proportional to a^4 ; the integrated finite-frame test then recovers general-triad scalar curvature with fitted power about 2.03. This is a constructive RG mechanism, not a uniqueness theorem for the final frame representation.

E11-E13 examine whether the enlarged representation leaves unwanted low-energy modes. Global transport shifts are pure gauge for difference-only Link sources, the BCC incidence operator has one exact collective null direction, and relative/frame excitation gaps separate parametrically from the collective mode in the tested blocks. The universality audit distinguishes relevant data - causal-cone speed and anisotropy, matter species/gaps, the Newton/weak-load normalization, DeWitt/HDA structure, and any ungapped frame modes - from $O(a^2 k^3)$ lattice corrections that vanish in the IR. Strong-load constitutive functions can agree in the weak sector while remaining physically different near saturation; therefore the IR universality class does not by itself select a unique fundamental cellular automaton.

30. Strong-gravity observability, angular momentum, and spin-squared response (E14-E18)

E14-E15 separate robust strong-gravity statements from observational overreach. For a Planck-scale regular core in a Hayward-type diagnostic exterior, ordinary shadow and eikonal-ringdown shifts scale as powers of the core-to-horizon ratio and are negligible for astrophysical masses. A near-horizon reflector would produce echo times in an observable band, but such a reflector is not part of the present GCC horizon construction; echoes therefore remain conditional. The finite density cap fixes a cube-root minimum core scale but does not choose bounce, stall, or mass-flux dynamics. Once an outer horizon has formed, strict locality prevents the saturated core from sending a new

classical signal through that horizon. These results disfavor treating the core itself as the generic source of observable merger-wave modifications.

Rotation likewise forces the geometry sector beyond a pointlike saturated core. A diagnostic bound $J \leq M R c$ shows that an astrophysical dimensionless spin cannot be stored solely in a Planck-sized core. E16 constructs exact finite angular-momentum exchange between matter and Link/frame records; on a periodic BCC graph the global vector record is conserved phase by phase, and the weak frame-dragging field converges to the transverse $J \times r/r^3$ form with $O(a^2)$ anisotropy error. E17 matches the asymptotic Kerr frame-drag coefficient $\kappa=2$ as an IR normalization, but lower-order mass and spin data do not determine the spin-squared quadrupole amplitude. E18 makes the STF spin-squared source finite and reciprocal; standard slow-rotation horizon regularity supplies an infrared Kerr selector, while the microscopic normalization and nonlinear rotating branch remain open.

31. Unitary no-hair and propagating finite TT radiation (E19-E22)

E19 demonstrates that local no-hair relaxation is compatible with exact finite unitarity. A localized anomalous quadrupole can transfer amplitude to outward frame/wave records under a time-independent nearest-neighbor unitary. On finite rings recurrence is unavoidable; before the return time the local survival probability decays while total norm is preserved. A spectral audit shows the precise failure mode: a stable bound eigenmode with finite defect overlap produces persistent hair, whereas a purely dispersive continuum permits local relaxation.

E20 then applies this criterion to the actual microscopic architecture and finds an important omission: the E12 frame scaffold has no vacuum inter-Link propagation gate, so an abstract radiation continuum cannot honestly be identified with the E12 spectrum. E21 fills this gap. Each physical TT polarization receives a finite Q/P qudit, onsite kinetic phases, and eight strict-local BCC matching-gradient phases. The linearized supercycle obeys the BCC graph-wave Floquet law and approaches $\omega=c|k|$ with $O(a^2)$ spatial-plus-splitting error at fixed CFL ratio. Exact finite-qudit diagonalization shows that the low quasienergy gaps converge to the same oscillator-like spectrum.

E22 removes an unnecessary phenomenological degree of freedom introduced by the earlier bound-mode diagnostic. In the minimal actual architecture, the spin-squared source is linear in the same TT field; it shifts the static equilibrium but does not create an independent defect oscillator. Completing the square gives a source-independent perturbation Hessian. A wrong quadrupole coefficient is therefore a deformation of the propagating TT field and can dephase into the wave sector. The remaining static freedom is the source normalization/strong-background equilibrium, not a separately postulated local no-hair oscillator.

32. Curved spin-2 sectors, finite constraint reduction, and the Schwarzschild chain (E23-E27)

E23 establishes that curved transport alone is not enough. Replacing the flat E21 gradient by a minimally coupled Schwarzschild scalar-like operator leaves a finite strong-field mismatch from the $l=2$ Zerilli potential under radial refinement. The missing information is the gauge/constraint reduction of the metric perturbation. Once the reduced Zerilli master operator is supplied, however, the same finite Q/P architecture carries it: nearest-neighbor radial gradient phases plus an onsite local curvature phase are exact finite unitaries, and the lowest radial spectrum converges at approximately second order. On the Schwarzschild branch the Zerilli coefficient is an algebraic function of the local lapse/load and areal-radius data, so no nonlocal mass label is required [19].

E24 constructs the finite reduction mechanism explicitly. Four finite canonical pairs can be mixed by an exact finite symplectic map, constrained and gauge-fixed to a two-dimensional phase space, and represented quantum mechanically by a rank-d physical projector inside d^4 configuration states. A radius-one centered stencil reconstructs the Zerilli-Moncrief coordinate with $O(h^2)$ error. Because the coordinate contains one derivative of a metric register, its finite resolution must scale one power faster: $\Delta K=O(h^3)$ is required to retain an $O(h^2)$ master-variable error, while algebraic registers may scale as $O(h^2)$. A nondynamical auxiliary constraint variable generates the missing even-parity onsite term through a local Schur complement, and the same elimination has an exact finite-field projector analogue.

E25-E27 then identify the physical curvature data. On the standard time-symmetric Schwarzschild slice $R^{(3)}=0$, but the orthonormal Ricci eigenvalues are nonzero: $R_{\hat{r}\hat{r}}=-2M/r^3$ and the two tangential eigenvalues are $+M/r^3$. The even constraint coefficient can therefore be written locally in terms of lapse and directional curvature. Locality, weak-field matching, endpoint behavior, and stability still allow a continuous family of strong-curvature corrections. The ambiguity is removed in the verified Schwarzschild linear sector by parity consistency. The covariant spin-2 curvature action on the sheet-TT tensor gives $3 R_{\hat{r}\hat{r}}$, so the odd potential is

$$U_{\text{odd}} = f [l(l+1)/r^2 + 3 R_{\hat{r}\hat{r}}] = f [l(l+1)/r^2 - 6M/r^3].$$

This is the Regge-Wheeler form [20]. Direct BCC holonomy reconstruction on the isotropic Schwarzschild slice drives the inferred $l=2$ coefficient to $k_{\text{eff}}=3.99963$ at $a=0.05$, with approximately $O(a^2)$ convergence. Requiring the independently derived odd potential and a minimal local rational even constraint sector to form a nondegenerate Darboux pair then selects

$$C_2(y) = 14 y (3y+8)/(3y+4)^2, \quad y=2M/r,$$

and the generated even potential agrees symbolically with the Zerilli potential. The same construction extends to all $l \geq 2$ and is consistent with the standard Regge-Wheeler/Zerilli Darboux relation [21]. This is a strong infrared selection chain rather than a derivation of the full nonlinear GCC constraint algebra.

33. Microscopic spin-2 curvature Hessian (E28)

E28 closes one more gap between continuum geometry and the finite wave carrier. The original E12 frame/source gate contains finite geometry, work, transport, and gauge-frame records but no independent TT amplitude pair, so a spin-2 Hessian cannot be obtained by Taylor expanding E12 alone. The DeWitt kinetic form resolves this cleanly: on the physical trace-free plus/cross plane, $\text{Tr}(\pi_{\text{TT}})=0$ and the generalized DeWitt metric reduces to a positive identity independent of the trace coefficient. The two E21 Q/P carriers are therefore the minimal physical kinetic extension.

More importantly, the factor 3 in the Schwarzschild spin-2 curvature term is not stored as a fitted constant. The full BCC-reconstructed orthonormal Riemann and Ricci tensors are contracted directly with the sheet-TT representation. The resulting two-by-two curvature Hessian becomes polarization independent and its mean eigenvalue divided by $R_{\text{rhat_rhat}}$ converges to 3 with fitted power 2.001. At $a=0.05$ the ratio is 3.00122 and the plus/cross eigenvalue split is at roundoff. A rotation of the TT basis transforms the Hessian covariantly to about 2.5×10^{-17} .

The reconstructed Hessian then controls an exact finite two-polarization Q/P onsite phase. With 21 states per polarization the local Hilbert sector has dimension 441; the unitary residual is about 1.8×10^{-13} , the first-excitation subspace capture exceeds 0.9999, and the local stiffness converges to the Regge-Wheeler value with $O(a^2)$ holonomy error. An autonomous 11-phase TT scheduler combines kinetic, BCC gradient, and curvature phases with exact inverse and zero scheduler leakage. The result is a constructive finite spin-2 gate, not a proof that this extension is the unique microscopic completion.

34. All-register finite-unitary existence and non-uniqueness (E29)

E29 assembles the previously separate finite sectors into one factorized 18-phase supercycle. The phases perform geometry-controlled matter evolution, reversible work computation, load/frame/transport backreaction, work uncomputation, capacity refresh, reversible loop-curvature caching, TT half-kinetic evolution, eight BCC matching-gradient layers, a curvature-controlled spin-2 phase, the second half-kinetic layer, and curvature-cache uncomputation. The formal local-patch Hilbert dimension is about 9.04×10^{13} , so a dense matrix is neither necessary nor practical; every factor is instead represented as a finite controlled unitary or permutation.

On an exact 32×7^4 superposed matter-plus-TT basis sector, the maximum norm drift through the full cycle is 2.8×10^{-15} , the work and curvature caches return to vacuum, and the explicit inverse recovers the initial amplitudes to about 5.0×10^{-15} together with all discrete control records. Four coherently superposed geometry/capacity/frame/loop sectors also return under the full inverse. The non-Abelian frame projector commutes with the source evolution to roundoff, and the original matter-backreaction identity survives tensor extension by the TT sector.

The curvature cache supplies a direct local bridge from finite transport Links to the spin-2 phase. For an oriented four-edge Z_7 loop, vertex gauge shifts cancel exactly in the loop sum; exhaustive and structured sweeps show no invariance or cache-inverse failures. Nevertheless E29 does not establish uniqueness. Two different symmetric splittings of the same finite Q/P Hamiltonian are exact unitaries, differ microscopically by $O(dt^3)$, and share the same $O(dt^2)$ continuum quasienergy. The correct v3.2 claim is therefore existence and compatibility of an all-register finite update, not a uniquely selected fundamental rule.

35. Integrated HDA and finite smeared deformation circuits (E30-E31)

E30 first tests the integrated canonical target after matter and spin-2 fields have been added. Matter, TT^+ , and TT_x Hamiltonians close on one total diffeomorphism constraint when they share the same inverse-metric structure function $\beta = h^{xx}$; the centered-difference H-H error scales as $dx^{1.99}$. Arbitrary ultralocal matter and curvature-controlled TT potentials cancel from the H-H bracket to roundoff. A 10 percent TT propagation-coefficient

mismatch instead leaves an approximately 8.5 percent closure floor, showing that a common causal/metric normalization is a nontrivial integrated condition.

The H-D test exposes a second necessary condition. If the geometry is frozen while $D[x_i]$ moves only matter/TT fields, the closure error approaches an approximately 8 percent floor. Adding the Lie drag of the spatial metric restores H-D convergence with fitted power about 1.99; D-D converges similarly. Joint finite-geometry refinement with $\Delta h = O(dx^2)$ preserves the full effective target, while fixed Δh leaves H-H/H-D floors. These results extend the earlier E5 renormalization lesson to the matter-plus-spin-2 system.

E31 then replaces effective Poisson evaluation by actual finite smeared circuit families. For smooth finite-dimensional canonical fields define $S_H[N, \epsilon] = \exp(\epsilon A_H[N])$ and $S_D[x_i, \epsilon] = \exp(\epsilon A_D[x_i])$. With ϵ proportional to dx^2 , the H-H group commutator approaches the diffeomorphism circuit on the matter and two TT sectors; the combined circuit error falls from 0.284 at $N=32$ to 5.97×10^{-3} at $N=256$. When the metric participates in the tangential circuit, H-D and D-D commutator errors scale approximately as dx^2 . Finite lapse/shift alphabets are also compatible provided their resolution is derivative-aware, $\Delta \text{smear} = O(dx^3)$; a fixed alphabet fails to approximate arbitrary continuum smearings under refinement.

The decisive negative result is the full-state H-H circuit. Two present normal-deformation circuits leave h unchanged, while the target diffeomorphism produced by their commutator Lie-drags h . The physical matter/TT error converges, but the full-state relative error remains about 78.8 percent in the finest tested case, almost entirely carried by the missing metric deformation. This is not treated as a numerical anomaly. It identifies the required microscopic register: a canonical geometry momentum/extrinsic-curvature sector. E31d constructs a finite diagonal DeWitt shear over F_{31} , $q \rightarrow q + n G p$ with $G = I - (1/2)11^T$, and verifies exact symplecticity and 536,238 forward/inverse state-lapse combinations without failure. A complete full H-H proof must still add the curvature momentum shear and the full frame/geometry diffeomorphism action.

36. Integrated conclusion of Version 3.2

Paper IV v3.2 retains the finite BCC substrate and one-law interpretation but materially changes the status of several previously open interfaces. Smooth frame holonomies now reconstruct directional curvature in explicit finite tests; finite physical TT registers propagate on the same graph; the reconstructed curvature tensor directly generates the spin-2 Hessian; and the Schwarzschild linearized branch closes from frame curvature to Regge-Wheeler and, by nondegenerate parity consistency, to Zerilli. These results do not establish nonlinear black-hole uniqueness, but they show that the physical spin-2 sector can be carried by finite local registers without inserting a continuum graviton equation as a separate microscopic law.

The integration status is also stronger. A concrete factorized all-register finite-unitary supercycle exists and remains reversible for coherent control sectors. Finite lapse/shift-dependent deformation-circuit families now exist as well, so HDA tests no longer rely only on inferring a Hamiltonian from one fixed Floquet unitary. Matter-plus-TT H-H, geometry-aware H-D, and D-D relations recover the intended continuum structure under joint refinement. The new negative control localizes the remaining covariance gap: normal deformations must act on the full geometry canonical pair, not merely on matter and TT amplitudes conditioned by a frozen frame record.

Accordingly, Version 3.2 still describes GCC as a candidate finite architecture. The highest-priority microscopic tasks are now: (i) extend the finite geometry register to the complete canonical frame/metric momentum sector and close the full smeared H-H circuit; (ii) select or derive the final finite frame/address representation and prove block-RG decoupling of extra physical modes; (iii) establish universal interacting matter and the stability/normalization of Newton coupling; (iv) derive nonlinear collapse and rotating black-hole branches rather than infer them from static or weak-spin diagnostics; and (v) calculate the exterior many-body scattering, greybody, and exclusive evaporation amplitudes. The existence results reported here substantially narrow these tasks but do not replace them by assumptions.

APPENDIX A. Exact finite-Weyl and BCC audit definitions

For a finite Weyl pair, matrix exactness can be monitored by the relative Frobenius residual

$$R_W = \frac{\| W_M W_N - \omega^k W_N W_M \|_F}{\| W_M W_N \|_F}$$

d	Weyl phase exponent	Relative matrix residual
5	2	2.18×10^{-16}

d	Weyl phase exponent	Relative matrix residual
7	4	5.75×10^{-16}
11	8	3.54×10^{-16}
17	14	4.31×10^{-16}

The reported values are floating-point roundoff, illustrating the distinction between exact finite matrix composition and emergent continuum closure.

APPENDIX B. Backreaction identity

Let the finite geometry shift advance the geometry basis by one record and let the matter update be diagonal in the geometry-control basis. Direct multiplication gives

$$C_{GM}^\dagger (X_G \otimes I) C_{GM} = X_G \Omega_G, \Omega_G = \sum_g |g\rangle \langle g| \otimes U_M(g+1)^\dagger U_M(g)$$

No measurement or classical stress extraction is used. In a smooth chart, expand the neighboring controlled unitaries to first order in the geometry-record step and identify the shift phase with the conjugate geometry momentum. The resulting momentum kick is the derivative of the same matter Hamiltonian that controls propagation. Higher-order Baker-Campbell-Hausdorff terms are ultraviolet corrections rather than independent source terms.

APPENDIX C. DeWitt one-half selection

For the general local quadratic kinetic term, only the kinetic-curvature cross bracket contributes second derivatives of the lapses. Contracting the curvature variation with the generalized momentum combination gives a standard momentum-constraint term and a trace-gradient remainder proportional to one minus twice the DeWitt coefficient. Antisymmetrizing the lapse pair and integrating by parts yields the expression in Section 10. If no additional trace constraint is admitted, standard HDA closure therefore selects one half. This is a conditional derivation within the local quadratic ansatz, not a global uniqueness theorem over arbitrary nonlocal or higher-derivative microscopic completions.

$$\{H_\lambda[N], H_\lambda[M]\} = D[\omega] + (1 - 2\lambda) \int d^3x (\nabla_a \pi) (N \nabla^a M - M \nabla^a N)$$

APPENDIX D. BCC loop curvature and strict locality

The BCC neighbor shell contains independent pairs of edge directions whose cross products span the spatial bivector space. Four-edge loops therefore provide enough independent holonomies to reconstruct the three spatial curvature two-form components in the smooth sector. A direct one-tick plaquette readout would violate radius-one causality because the opposite loop corner is not in the node neighborhood. The reversible scheduler instead transports and multiplies one edge factor per microscopic tick. A completed loop record is cached only after all factors have arrived. The finite delay is derivative suppressed in the infrared and becomes a genuine ultraviolet memory effect near the processing scale.

$$W_{x;s,t} = L_{x,s} L_{x+s,t} L_{x+t,s}^\dagger L_{x,t}^\dagger$$

$$W_{x;s,t} = \exp \left[-\frac{i}{4} \Sigma_{s,t}^{ab} F_{ab}^i \sigma_i + O(\ell_{proc}^3) \right]$$

APPENDIX E. Correlated strong-gravity radial equation

Using the spherical mean-density load, mass conservation and the explicit Compute-Link circuit imply

$$r \frac{d\chi}{dr} = -3\chi(1-\chi) \cos^2(g\chi^2)$$

The uncorrelated limit is logistic and gives the analytic Hayward-type branch. For finite correlation the equation is separable in principle but not generally elementary. The endpoints remain fixed because the circuit correction vanishes at zero and unit load. The weak-load expansion preserves the quadratic leading term, while the correlation

correction starts at fifth order. Hence the central density cap and cube-root mass scaling are more robust than the detailed exterior coefficient.

APPENDIX F. Physical-state construction at microscopic level

The microscopic gauge problem is separated from the emergent Hamiltonian constraint problem. Internal frame rotations and address relabelings act as finite redundancies of the record description. Projecting onto their invariant sector gives a candidate microscopic physical Hilbert space.

$$P_{phys} = P_{addr} P_{frame}, H_{phys}^{micro} = P_{phys} H, [U_{GCC}^{full}, P_{phys}] = 0$$

A nonzero invariant vacuum guarantees that the projected space is not empty. The E8 Z3 plaquette model and E9 binary-tetrahedral 2T model admit explicit gauge projectors, gauge-invariant curvature observables, and finite evolution that preserves the projected sector. Version 3.2 adds a gauge-invariant finite loop-curvature cache and shows that the projected frame sector remains compatible with the all-register TT extension. The actual GCC frame/address representation remains to be selected, and physical frame excitations must still be shown to decouple dynamically. What has changed is the curvature status: a smooth inhomogeneous frame-holonomy-to-curvature reconstruction is now constructive in the tested backgrounds, although its uniqueness and full nonlinear embedding are not yet established.

APPENDIX G. Horizon dimension matching and information transfer

The binary boundary algebra implies exponential horizon information growth with the number of independent records. Removing a finite number of records while storing the same records in radiation preserves the total Hilbert-space dimension exactly.

$$\dim H_H(n) = 2^n = 2^{n-q} 2^q = \dim [H_H(n-q) \otimes H_R(q)]$$

This equality is the algebraic reason a lossless finite-dimensional transfer is possible. The explicit map in Section 21 is a convenient bijection; the full interacting QCA may realize a more complicated energy- and channel-dependent unitary within the same dimension-matched block. Microcanonical reciprocity depends only on reversibility and state counts, so the inclusive normalization survives that microscopic freedom.

APPENDIX H. Logical hierarchy of the v3.2 construction

Level	Exact / finite requirement	Emergent / conditional requirement
Microscopic algebra	Finite matrices; exact unitarity; finite Weyl composition	Canonical variables only in low-phase charts
Microscopic causality	One-neighborhood support per microscopic scheduler phase	Continuum light cone in low-wave-number sector
Gauge	Finite frame/address covariance; explicit finite projectors in prototypes	Final non-Abelian frame/address representation and complete physical-state sector
Matter	4D exact flat reference; 32D finite curved-carrier candidate	Curved Weyl/Dirac QFT and universal interacting continuum fields
Gravity	Bounded finite-Weyl geometry, full smooth BCC frame-curvature reconstruction, finite TT Q/P carriers, and curvature-controlled spin-2 gates	ADM/DeWitt universality, full geometry-inclusive HDA, and Einstein-Hilbert action
Renormalization	Reversible block variables; exact collective eta,zeta -> 1/B	Closed interacting RG with dynamic decoupling of relative/frame modes
Frame curvature	Gauge-covariant finite holonomies, reversible loop caches, directional Ricci reconstruction, and spin-2 tensor Hessian	Unique final frame/address representation and nonlinear all-background curvature dictionary
Strong gravity	Finite capacity and bounded constitutive response	Dynamical collapse, rotation, and stability of the regular branch
Horizon information	Rank-two outgoing carrier algebra; finite transfer bijection	Boundary-support minimality and area matching
Evaporation	Finite unitary block transfer; microcanonical reciprocity	Greybody spectrum, line broadening, full interacting Page dynamics
Spin-2 sector	Finite two-polarization Q/P registers; strict-local BCC wave scheduler; finite RW/Zerilli master-sector realizations	Universal nonlinear graviton sector, Kerr perturbations, and strong-background stability
All-register update	Exact factorized 18-phase finite-unitary	Microscopic uniqueness / universality of

Level	Exact / finite requirement	Emergent / conditional requirement
	existence prototype with coherent-control inverse tests	update ordering
Smeared deformations	Finite lapse/shift-dependent $H[N], D[xi]$ circuit families; physical H-H and geometry-aware H-D/D-D group-commutator tests	Complete geometry-momentum/curvature normal deformation and full operator-level microscopic HDA

APPENDIX I. Selected v3.2 numerical audit values

- E10: general-triad/finite-frame integrated curvature error fits approximately a^2 ; one representative integrated fit is $a^{2.03}$. A fixed 2T/T weak-link alphabet fails, while the 2T-gauge x near-identity transport representation succeeds under RG-refined transport resolution.
- E21: the actual BCC Floquet eigenphases agree with the analytic graph-mode law to numerical precision on finite periodic tests; fixed-CFL continuum dispersion approaches $\omega = c|k|$ with approximately $O(a^2)$ directional error.
- E27: on the isotropic Schwarzschild slice the reconstructed odd coefficient reaches $k_{\text{eff}} = 3.99963$ at $a = 0.05$, with $|k_{\text{eff}} - 4|$ fitting approximately a^2 .
- E28: the direct TT curvature-Hessian eigenvalue ratio reaches 3.00122 at $a = 0.05$; full target error scales approximately a^2 and polarization splitting is at roundoff.
- E29: the 18-phase integrated basis-control sector has maximum norm residual 2.8×10^{-15} , full inverse residual 5.0×10^{-15} , and zero work/cache leakage; coherent four-control-sector inversion is 4.4×10^{-15} .
- E30: common-metric matter+TT H-H closure scales as $dx^{1.99}$; a 10 percent TT gradient mismatch leaves an 8.5 percent floor. Geometry-aware H-D and D-D converge approximately as dx^2 .
- E31: physical H-H circuit error falls to 5.97×10^{-3} at $N = 256$; H-D and D-D circuit fits are approximately $dx^{2.00}$ and $dx^{1.95}$. Full-state H-H remains about 78.8 percent because the target deformation is dominated by the metric motion absent from the current H circuit.

APPENDIX J. Schwarzschild spin-2 local chain

On the time-symmetric Schwarzschild spatial slice, the scalar curvature vanishes while the directional Ricci curvature does not. In an orthonormal frame, $E = R_{\hat{r}\hat{r}} = -2M/r^3$ and the tangential eigenvalues are $-E/2$. For a transverse-traceless tensor on the angular sheet, the local three-dimensional curvature endomorphism gives $K_{\text{curv}}(W) = 3E W$. The local odd-parity master coefficient is therefore fixed by tensor geometry rather than by a fitted wave potential.

$$U_{RW} = f [l(l+1)/r^2 + 3E] = f [l(l+1)/r^2 - 6M/r^3].$$

For $l=2$, taking this odd sector as independently fixed and imposing the nondegenerate Darboux relation on the minimal local even constraint class selects $C_2(y) = 14y(3y+8)/(3y+4)^2$ and reproduces the standard Zerilli potential. The cited Schwarzschild perturbation literature supplies the continuum comparison only; the BCC curvature reconstruction and finite gate tests are the GCC-specific numerical steps [19-22].

APPENDIX K. E29 integrated scheduler

- 0: geometry-controlled matter update.
- 1: reversible geometry-difference work compute.
- 2: load/frame/transport reciprocal source update.
- 3: work uncompute.
- 4: capacity-conditioned geometry refresh.
- 5: oriented Link-loop curvature cache compute.
- 6: TT half-kinetic phase.
- 7-14: eight BCC perfect-matching TT gradient phases.
- 15: curvature-controlled spin-2 TT phase.
- 16: TT half-kinetic phase.
- 17: curvature-cache uncompute.

Every listed phase is finite and invertible. Strict microscopic locality is implemented by the matching/scheduler decomposition; the entire loop curvature need not be readable in one graph tick.

APPENDIX L. Smeared-deformation stopping point

A finite family of deformation circuits is stronger than a single Floquet unitary because the smearing data are explicit inputs. The v3.2 tests use small group commutators to extract the local algebra. The verified physical-sector relations are consistent with

$$\begin{aligned} [H[N], H[M]] &\rightarrow D[h^i_j (N d_j M - M d_j N)], \\ [H[N], D[xi]] &\rightarrow H[-L_{xi} N], \quad [D[xi], D[eta]] \rightarrow D[[xi, eta]], \end{aligned}$$

up to the convention-dependent sign of the generators and the stated refinement errors. The present full-state failure is equally informative: without a geometry momentum/extrinsic-curvature action in $H[N]$, the left-hand side cannot move h_{ij} whereas the target D circuit does. E31d supplies a finite DeWitt kinetic shear as a constructive register-level fix, but the curvature momentum update and complete geometry diffeomorphism sector have not yet been assembled into the all-register circuit. Hence operator-level microscopic HDA remains open in v3.2.

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